Module 3: Problem Discovery and Market Research

Find problems worth solving: research methods, interviews and surveys, personas, jobs-to-be-done, journeys, synthesis and evidence quality.

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MODULE 3

Problem Discovery and Market Research

Find problems worth solving: research methods, interviews and surveys, personas, jobs-to-be-done, journeys, synthesis and evidence quality.

≈ 10 h study

About examples

Personas, quotes, review counts and estimates are **invented** to teach technique. They are not research findings about any real product or user group.

3.1 What discovery is

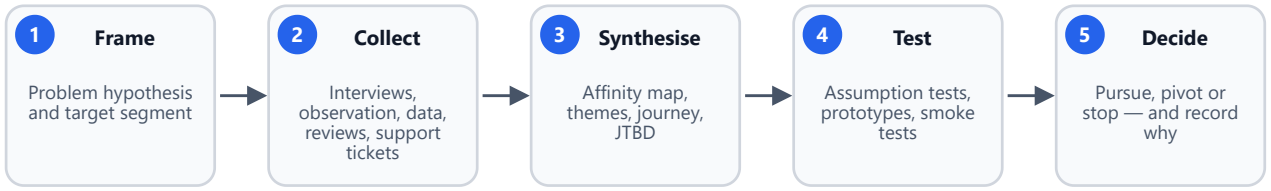
Product discovery is the work of learning *what is worth building* before (and while) building it: who has the problem, how painful it is, how they cope today, and whether your idea would be chosen and used. **Customer discovery** focuses on understanding customers; **problem validation** tests that the problem is real, frequent and painful enough to matter.

Level 1 · Beginner

Discovery = talking to people and watching what they do, so you build the right thing. If you can only ask “Would you use my idea?”, you will mostly hear polite yeses.

Key points: validate before you build

Confirm the problem before committing serious money to a solution: is the need real, who has it, how do they cope today, and is it worth solving? Any solution is a bet to be tested. Begin with a specific user and task, ask about a recent real occasion, observe the task when you can, and look for difficulties that repeat. People saying an idea “sounds nice” tells you very little.



A discovery cycle. Discovery repeats; each loop is cheap relative to building. Most teams under-invest in Synthesise and Decide.

3.2 Research methods: qualitative and quantitative, primary and secondary

Two different axes — *do not confuse them*:

	Qualitative (why / how)	**Quantitative (how many / how much)**
Primary (you collect it)	Interviews, observation, diary studies, usability tests, moderated focus groups	Surveys, A/B tests, instrumented behaviour in your own product
Secondary (already exists)	Support tickets, app-store reviews, forums, published case studies	Market reports, public statistics, company databases, benchmarks

Key points: kinds of evidence

Qualitative versus quantitative describes the type of evidence; primary versus secondary describes who collected it. An interview is primary and qualitative; a well-designed survey can be primary and quantitative. A focus group is a facilitated discussion, not a synonym for target audience. A handful of interviews points to problems but cannot tell you how common they are across a whole population.

Question you have	Best-fit method	What you get	What you don't get
Why do users abandon checkout?	Interviews + session observation	Reasons, context, language	How common each reason is
How many abandon at the fee step?	Funnel analytics	Counts and rates	Why
Which of three designs is clearer?	Usability test (5–8 per round)	Where people struggle	Statistical proof
Which price converts better?	A/B or priced pilot	Behaviour at scale	Reasons behind it
How big is the segment?	Bottom-up sizing + survey	An estimate with assumptions	Proof of demand
What do users say publicly about rivals?	Review mining	Themes and vocabulary	Representative opinion

Choosing a method by question. Mix methods: use analytics to find *where*, interviews to learn *why*, experiments to test *whether the fix works* (triangulation).

3.3 Customer interviews

3.3.1 Principles

1. **Ask about the past, not the future.** “Tell me about the last time you ordered lunch” beats “Would you use an app that…?”
2. **Seek stories, then specifics:** what triggered it, what alternatives they considered, what happened, how they felt, what they did next.
3. **Stay neutral:** don't pitch; don't lead; tolerate silence.
4. **Recruit for contrast:** include people who tried and churned, people who never tried, power users — not only fans.
5. **Consent and privacy:** explain purpose, record only with permission, remove identifying details in notes.
6. **Debrief fast:** write observations separately from interpretations within 24 hours.

Key points: interviewing well

Example: ask someone who buys lunch at work to describe the last time they ordered it. If they mention being surprised by fees, watch where on the screen they expected the price to appear. Small interview sets show problems, not their frequency. Recruit a mix — including people who gave up as well as those who finished — get consent, keep questions neutral, strip identifying details from notes and cross-check themes against usage data.

3.3.2 Question design

Weak (leading / hypothetical)	Stronger (past behaviour, neutral)
"Would you pay for faster delivery?"	"Last time you were in a hurry, what did you do about food?"
"Do you find our checkout confusing?"	"Walk me through placing your last order. What did you look at first?"
"Don't you hate hidden fees?"	"How did you decide whether the total was acceptable?"
"Is price important to you?"	"Tell me about a time you decided *not* to order because of the cost."
"What features do you want?"	"What did you do the last time this didn't work?"

3.3.3 Sample interview script (45 minutes)

Script: ordering food after work (hypothetical study)

Opening (5 min). "Thanks for your time. I'm learning how people decide about dinner on weekdays. There are no right answers; I'm not testing you. May I record audio for my notes only? I'll remove names." **Warm-up (5 min).** "Tell me about your typical weekday evening." **Core story (20 min).** "Think about the last time you ordered food from an app on a weekday. What was happening before you opened it?" → "What did you do first?" → "What else did you consider?" → "What were you looking at when you decided?" → "Was there a moment you almost gave up?" → "What did you do next?" **Probe the friction (10 min).** "You mentioned the total changed at the end — what did you expect to see?" → "How did that feel?" → "What did you do?" **Wrap-up (5 min).** "Is there anything I should have asked?" → "Who else should I speak to?" → thank, explain next steps. **Don't:** pitch your idea, ask about features, or suggest answers.

3.3.4 Recording the evidence

Use a two-column note sheet: **Observation** (what they did/said, verbatim where possible) | **Interpretation** (your hypothesis). Tag each observation with the participant ID and the journey stage.

3.4 Surveys and questionnaire bias

A **survey** estimates *how common* something is — if the sample is representative and the questions are neutral.

Common biases and fixes

Bias	What happens	Fix
Leading question	Wording nudges the answer ("How great was…?")	Neutral wording; balanced scales
Double-barrelled	Two questions in one ("fast and cheap?")	One idea per question
Social desirability	People give flattering answers	Anonymity; behaviour ("last week, how many times…?")
Acquiescence	Tendency to agree	Mix positively and negatively worded items; use specific choices
Order effects	Earlier questions prime later ones	Randomise; put sensitive items last
Sampling / self-selection	Respondents differ from the population	Define the frame; track response rate; compare with known figures
Hypothetical bias	Stated intent ≠ behaviour	Prefer behavioural questions; validate with a real action
Incentive bias	Rewards attract the wrong people	Screen carefully; use modest incentives

Key points: surveys

Surveys are not inherently biased; the risk comes from who is sampled, how they are rewarded and how questions are worded. And a stated preference in a survey is not evidence that people will buy.

Sample survey questions (rewritten)

- Bad: “Do you agree our app is easy and fast to use?” → Good: “In the past 7 days, how many times did you place an order? (0 / 1 / 2–3 / 4+)” and “How easy or difficult was it to find the total price on your last order? (Very difficult … Very easy)”.
- Bad: “Would you use a feature that tells you about fees?” → Good: “When you last saw a price at checkout, was the total different from what you expected? (Yes, higher / Yes, lower / No / Don't remember)”.

3.5 Customer segmentation and personas

A **segment** is a group sharing a *relevant* need or behaviour. A **persona** is an illustrative profile of a segment — a communication aid, not proof everyone acts alike. Prefer **behavioural and need-based** segments over purely demographic ones.

Key points: audiences and personas

A target audience is the group a product or message is meant to reach. The purchaser and the user can be different people — think of a parent buying something a child will use — and both views matter. Personas should come from research, not imagination, and there is no fixed number every product must have. Picking the biggest segment does not prove fit. When segments want different things, weigh size, how painful the need is, strategic fit, feasibility and how easily you can reach them; some overlap between personas is normal.

NAME & CONTEXT

- "Asha", 31, product analyst
- Works late 3–4 days a week
- Uses food apps on her commute

GOALS & MOTIVATIONS

- A reliable dinner without planning
- Keep weekday spend predictable

BEHAVIOURS (FROM INTERVIEWS)

- Re-orders favourites; checks total before paying
- Abandons if total jumps at last step

PAINS / FRUSTRATIONS

- Fees appear late
- Delivery estimates drift
- Unclear refund path

NEEDS FROM THE PRODUCT

- Clear total early
- Trustworthy ETA
- One-tap reorder

EVIDENCE & CONFIDENCE

- 8 interviews (Mar 2026, hypothetical)
- Confidence: medium
- Open questions: what happens at month-end?

Persona template filled with a hypothetical example. A persona is only as good as its evidence. Always record sample size, date and confidence, and update it when research changes.

3.6 Jobs to Be Done (JTBD)

JTBD frames demand as **people "hiring" a product to make progress in a particular circumstance.** A **job statement** has the form:

```
When [situation], I want to [motivation / progress], so I can [desired outcome].
```

- Jobs have **functional**, **emotional** and **social** dimensions.
- The *competition* is whatever else gets hired for the job — including doing nothing.
- Uncover jobs from **switching stories**: what triggered the move, alternatives considered, what pushed them away from the old solution (push), what pulled them to the new one (pull), what anxieties held them back, what habits kept them in place — the **four forces**.

Example (hypothetical): lunch

When I'm in back-to-back meetings and have 15 minutes, I want to get a filling lunch without leaving my desk or being surprised by the cost, so I can stay focused and not blow my budget. **Competing alternatives:** order in, ask a colleague to pick up, eat a snack, skip lunch, bring food. **Anxieties:** "Will it arrive in time? Will the price change?" **Implication:** reliability and total price clarity matter more than restaurant variety for this job.

PUSH (AWAY FROM THE CURRENT WAY)

- Wasted groceries
- Planning takes an hour

PULL (TOWARD THE NEW SOLUTION)

- Weekly plan with a price estimate

ANXIETY (ABOUT THE NEW SOLUTION)

- Will estimates be accurate? Will it fit our diet?

HABIT (OF THE PRESENT WAY)

- Family's recipe notebook; saved lists in notes app

The four forces of switching (hypothetical switch to a new meal-planning app). A product wins when push + pull outweigh anxiety + habit. Discovery asks about each force; design and onboarding reduce anxiety.

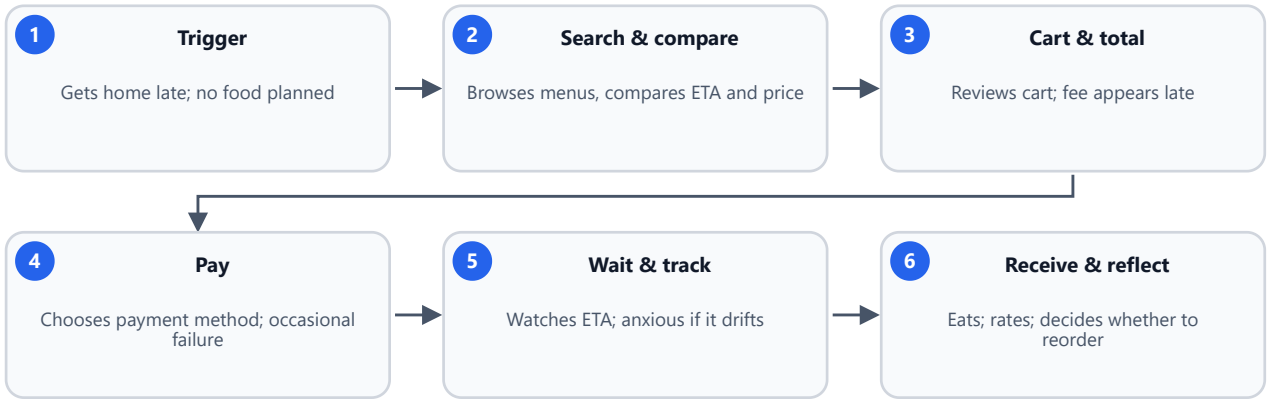
3.7 Pain-point identification

List pains with **severity** (how bad), **frequency** (how often), **breadth** (how many people) and **current workaround** (the clearest sign of real pain: people already spend effort to solve it). Rank by severity × frequency × breadth, then by whether you can credibly solve it.

3.8 Customer journey mapping and user flows

Key points: journeys, flows and workflows

A journey map covers the whole experience — actions, thoughts and feelings across every touchpoint. A user flow zooms into the paths and decisions within the product, and one journey contains many flows. A workflow tracks the sequence of activities, systems and decisions behind the scenes. Start from a concrete task, map the steps and the branches that matter, and pick a level of detail that fits the question. AI can sketch a first draft if you give it the problem, users, features and constraints, but compare it with the real product before using it. Make sure the important branches — bad address, failed payment, retry, cancellation — each have a defined outcome. A remembered login is not automatically single sign-on.



Journey: weekday dinner order (hypothetical, one persona). Journey stages are the rows of a map; add rows for actions, thoughts, emotions, touchpoints, pain points and opportunities.



Emotion curve across the journey (hypothetical, from 8 interviews). Hypothetical. The two lowest points — the cart total and payment — point to where discovery and design effort should focus.

3.8.1 Opportunity Solution Trees (OST)

Teresa Torres's **Opportunity Solution Tree** connects a **desired outcome** to the customer **opportunities** (needs, pains, desires) and then to **solutions** and **experiments**. It prevents jumping from outcome straight to a favourite feature.



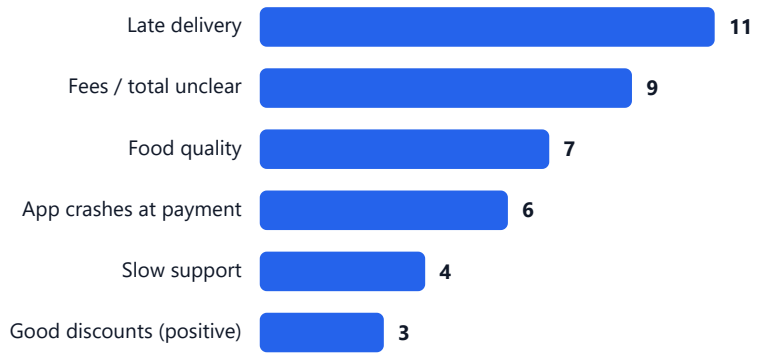
Opportunity Solution Tree (hypothetical). Each solution should be paired with the cheapest experiment that tests it. Compare opportunities (not just solutions) when you prioritise.

3.9 Competitor research, review mining and support-ticket analysis

Review mining turns public reviews and support tickets into themes and a vocabulary. **Method:** (1) sample reviews across ratings and dates; (2) code each into themes (one review can have several); (3) count and read representative quotes; (4) note sampling limits; (5) turn themes into hypotheses.

Worked example — coding 40 synthetic app reviews

Theme	Mentions	Share of 40 reviews
Late delivery	11	27.5%
Fees / total unclear	9	22.5%
Food quality	7	17.5%
App crashes at payment	6	15.0%
Slow support	4	10.0%
Good discounts (positive)	3	7.5%



Themes in 40 synthetic reviews (hypothetical). Synthetic data for teaching coding. With 40 reviews you can form hypotheses, not estimate the share of all customers. Reviews over-represent extreme experiences.

Key points: mining reviews with AI

AI can help tag sentiment and themes such as reliability, price, safety and support in a set of public reviews, but read individual comments and think about who writes reviews; a small sample cannot represent the whole market. Check where any public dataset came from and under what licence — it may be real, synthetic or both. Never upload confidential material without permission.

3.10 Behavioural analytics and observation

- **Behavioural analytics** (funnels, cohorts, session replays where permitted) shows *what* people do at scale (Module 7).
- **Observational research** — watching users perform tasks (in context if possible) reveals what they cannot articulate: workarounds, hesitations, misreadings. Think-aloud usability testing is a form of observation.
- Always combine: analytics finds *where*; observation and interviews find *why*.

3.11 Research synthesis

3.11.1 Affinity mapping

Write each observation on a note; cluster by similarity (silently first, then discuss); name clusters as **insights** (“Users distrust totals that change at the end”), not topics (“Pricing”); count participants per cluster.

INSIGHT A · 6 OF 8

- "The total jumped at the end."
- "I didn't see delivery fee until payment."
- "I wasn't sure what I was paying."

INSIGHT B · 4 OF 8

- "If payment fails I start over."
- "I switched to cash once."

INSIGHT C · 3 OF 8

- "I just want to reorder my usual."
- "Saved addresses save me time."

Affinity map output (hypothetical, 8 participants). Hypothetical. Report counts as "6 of 8 participants", never percentages from a handful of interviews.

3.11.2 Assumption mapping and hypothesis generation

List every assumption behind the idea (desirability, usability, feasibility, viability). Plot on **importance** (how fatal if wrong) vs **evidence** (how much we know). Test the **important, low-evidence** assumptions first.



Assumption map (hypothetical meal-planning app). Hypothetical. Top-left (important, little evidence) is where experiments should start.

Hypothesis template: *We believe that [change/solution] for [segment] will [expected outcome/metric] because [reason/evidence]. We will know we are right when [measurable signal].*

3.12 Problem validation and evidence quality

Triangulation means reaching a conclusion from multiple independent kinds of evidence.
Grade evidence honestly:

Strength	Evidence	Example
Strong	Observed behaviour at scale; randomised experiments; paid pilots	Conversion lift in an A/B test; people paying a real price
Medium	Repeated behaviour in small samples; consistent interview themes; usability test patterns	6 of 8 users miss the fee in a task
Weak	Stated intent, opinions, single anecdotes, AI-generated claims, "I would use this"	Survey says 80% "would try"

> **Key points: grading evidence honestly**
>
> Call comparisons based on personal impressions what they are: hypotheses. Support important claims with sources and a stated method. Numbers produced by an AI tool are not evidence until you can trace them. Your own experience is a useful starting point but does not prove a problem is widespread.

3.13 Guesstimates (estimation)

A **guesstimate** breaks an unknown number into smaller numbers you can explain, state assumptions for, and sanity-check — and gives a **range**.

> **Key points: estimation questions**
>
> First agree the scope — area, an ordinary day or a special event, which product types, whether cancelled requests count — and keep those definitions consistent. Break the question into assumptions you can explain, sanity-check each, and compare a top-down with a bottom-up estimate. Give a range and name the assumptions that move it most. There is no single correct number, but maths errors, shifting definitions or unrealistic assumptions make any estimate weak. Tiny example: 1,000 pupils × 60% present × 50% buying = 300 lunches.

Worked example — daily ride-hailing trips in a large metro (practice numbers, not facts)

Top-down: metro population 20 million → adults 70% = 14.0 million → people who use ride-hailing at least weekly: 8% = 1.12 million → 3 trips per week each = 3.36 million trips/week → **≈ 480,000 trips/day. Bottom-up (supply check):** active drivers per day 55,000 × 9 completed trips per driver per day = **≈ 495,000 trips/day. Reconcile:** two independent approaches land within ~3% — *this raises confidence but both rest on assumptions I invented.* Give a range: **450k–500k/day**, and name the most uncertain assumptions (share of weekly users, trips per driver). Sanity check: 480k trips/day ÷ 14M adults ≈ 3.4% of adults ride each day — plausible for a big city.

Structure to use in interviews: clarify → choose approach (top-down or bottom-up) → segment → state assumptions → compute → sanity-check → give range → discuss what would change it.

3.14 Product critique and teardown

A **teardown** studies a product from the outside to understand users, goal, journey, strengths and possible improvements. Separate **observations** from **assumptions**; an outsider cannot know a company's internal data.

Key points: critiques and teardowns

State the user, the task, the friction and the benefit of fixing it. A workable teardown order: context and problem → purpose → personas → user and market research → journey and pain points → prioritised opportunities → proposed solution → success metrics; adapt it to the question. Every claim needs evidence and every proposed change needs a test. Improvement ideas you form from the outside are proposals, not proven gaps, and constraints such as content licences and contracts may explain why a product works the way it does.

Teardown skeleton (use for any product)

1. **Product & user:** what job; for whom; business model (assumption, label it).
2. **Journey:** 5–7 stages; where does it help; where does it hurt (evidence: your own walk-through, reviews, tests).
3. **Opportunity:** one problem; evidence; who is affected; size (assumption).
4. **Solutions:** 2–3 options incl. a non-AI/simple one.
5. **Choice:** impact/effort/risk; recommendation.
6. **Measurement:** primary metric; guardrails; experiment.
7. **Risks and unknowns.**

3.15 Discovery exercises

Exercise 1 · Rewrite the questions

Rewrite three poor interview questions and explain the bias in each: (a) “Do you think our onboarding is too long?” (b) “Would you pay ₹99/month for this?” (c) “What features would make you love this app?”

Model answer

(a) Leading and closed → “Walk me through your first time setting up the app. Where did you hesitate?” (b) Hypothetical bias → “What do you pay for today to solve this? How did you decide it was worth it?” (c) Solution-seeking; customers aren't product designers → “Tell me about the last time this task went badly. What did you do?”

Exercise 2 · Evidence grading

You have: (1) 8 interviews, 6 mention surprise at fees; (2) a funnel showing a 30% drop at the fee step; (3) a survey where 78% say they “like transparent pricing”. Rank the evidence and state what you would do next.

Model answer

(2) funnel = behavioural, scalable (strong *that* a drop exists, not why). (1) interviews = medium (consistent reasons, small sample). (3) survey = weak (stated preference, hypothetical). Next: segment the funnel (device, new/returning), run a usability task on the fee screen, then test early fee display in an A/B experiment with a complaint guardrail.

Exercise 3 · Estimate

Estimate how many cups of coffee are sold in a city of 5 million in a day. Show two methods and a range.

Model answer (example)

Top-down: 5M × 70% adults = 3.5M; 55% drink coffee ≈ 1.9M; 1.8 cups/day from cafés/vending/home? Only 35% bought outside → 1.9M × 1.8 × 0.35 ≈ **1.2M purchased cups/day**. **Bottom-up:** 2,500 cafés × 250 cups = 625k + 1,500 kiosks × 300 = 450k + vending/office 100k ≈ **1.18M**. Range 0.9–1.4M. Sensitive to share who buy outside and cups per café.

3.16 Common misconceptions

- Qualitative/quantitative ≠ primary/secondary.
- Focus group ≠ target audience.
- Persona count is not fixed; personas must be evidence-based.
- "Surveys don't work in some country" is unsupported; design matters.
- AI summaries of research need source checking.

3.17 Interview perspective

"How would you validate a problem before building?" — 60-second answer

"I'd start with a clear problem hypothesis and target segment, then interview 8–10 people who've recently experienced it, asking about their last real experience and what they do today. I'd look at available data to size and locate the problem, mine reviews or support tickets for patterns, and write down the assumptions that could kill the idea. For the riskiest assumption I'd run the cheapest test — a prototype or a smoke test with a real price — and decide in advance what result would make me continue, pivot or stop."

"Tell me about a product you love and how you'd improve it."

Tests: product sense, empathy, prioritisation, honesty about evidence. *Structure:* user + job → why it works → one meaningful pain (with evidence you'd gather) → 2–3 solutions → pick one with trade-offs → metric + guardrail + experiment. *Mistakes:* naming five features, ignoring constraints (content rights, privacy, cost), assuming your own experience is universal.

3.18 Key takeaways

- Discovery reduces **risk** by learning *before* building. Treat solutions as hypotheses.

- Ask about **past behaviour**; recruit for contrast; separate observation from interpretation.
- Match method to question; **triangulate** (analytics → interviews → experiment).
- Surveys estimate frequency but suffer from bias — design carefully.
- Use **JTBD**, journey maps, OSTs and assumption maps to structure what you learn.
- Grade evidence honestly; label hypotheses and AI-generated claims.

3.19 Check your understanding

CHECK YOUR UNDERSTANDING

Which is an example of secondary quantitative research?

A. A survey you design and send

B. A published market report with spending figures

C. Five customer interviews

D. A usability test

Why: Secondary research uses existing information; quantitative means numeric evidence. A published report fits both.

Which interview question is best?

A. Would you use an app that shows fees early?

B. Don't you hate surprise fees?

C. Tell me about the last time the total changed at checkout.

D. Which features do you want?

Why: It asks about a specific past experience, is neutral and invites a story.

In the four forces of switching, 'anxiety' refers to…

A. Frustration with the current solution

B. Attraction to the new solution

C. Worries about trying the new solution

D. Inertia of habit

Why: Push and pull move people toward switching; anxiety and habit hold them back.

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6 of 8 interviewees mention the fee surprise. How should you report it?

A. 75% of customers dislike hidden fees

B. 6 of 8 interviewed participants mentioned it; frequency in the population is unknown

C. It's statistically significant

D. Ignore it; the sample is too small

Why: Small qualitative samples reveal themes, not population percentages.

On an assumption map, which should you test first?

A. High importance, low evidence

B. Low importance, high evidence

C. Low importance, low evidence

D. High importance, high evidence

Why: Assumptions that would sink the idea and that you know little about carry the most risk per test.

A guesstimate gives 480k/day top-down and 495k/day bottom-up. The right conclusion is…

A. The answer is exactly 487,500

B. Confidence is higher but both depend on assumptions; give a range

C. Both are wrong

D. Use the larger number

Why: Agreement of independent methods raises confidence, but communicate a range and the key assumptions.

3.20 Recommended resources

- Nielsen Norman Group: Heuristic evaluations and user interviews.
- Product Talk: Opportunity Solution Trees.
- Rob Fitzpatrick, *The Mom Test* — asking good questions about past behaviour (book site).
- Jobs to Be Done — Christensen Institute.
- Google Cloud / Google: Gen AI glossary for AI-assisted research caveats.